Calibrating Interpretability Instruments Before Trusting Their Verdicts

Orion Reblitz-Richardson*

Abstract

Causal claims about language-model internals rest on measurements: a projection, a cosine, an ablation delta, an interchange patch. These measurements fail in specific, diagnosable ways that return a plausible number instead of an error, so a broken instrument reads as a finding. A covariance-matched null can saturate until every direction looks typical; a per-head attribution can overshoot the true residual write threefold on a common architecture family; an interchange patch can go sign-chaotic because its outcome is pinned at a ceiling; a read-from verdict can be an artifact of measuring past the layer where the model already decided.

This note documents six such failures from a causal interpretability program on refusal and moral representation across four open-weight architectures. For each we give the tell that catches it and a protocol keyed to a detectable trigger (reordered normalization, massive activations, a low-dimensional decision channel), so readers can check whether their own setup is exposed. The discipline reduces to four moves: calibrate against a positive-control ladder, certify with an orthogonal cell, compute power before spending compute, and state every read-from verdict at a depth referenced to the model's commitment. Applied to the program's own published paper, the discipline caught an abstract-level error. The evidence is four architectures across three families within a single program; external replication across programs is future work. A causal claim is only as good as the instrument behind it.

1 Introduction

A causal interpretability program kept producing results that dissolved under an instrument check. Each of the failures below looked like a finding first. This note collects the six that turned into portable methods findings and the estimator and intervention patterns the program re-derived, and states each as a protocol we found portable within this program and offer for others to test. The scientific results (what refusal reads, how it commits) live in the direction papers and the flagship draft; this note is the portable methodology. Numbers here trace to the program's claim ledger; every scalar carries its detection bar or its control.

The discipline is four moves: **calibrate the instrument against a positive-control ladder, certify it with an orthogonal cell, compute power before spending compute, and state every read-from verdict at a stated depth relative to the model's commitment.** Each section below takes one instrument, shows the failure as it first appeared, names the tell that caught it, gives the protocol, and states the check that certifies the fix.

Notation and definitions. The *participation ratio* of a position is $\mathrm{PR} = (\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$, where the λ_i are the eigenvalues of the residual-stream covariance at that position; it is an effective-dimension estimate of that covariance, equal to the true dimension for an isotropic covariance and dropping toward 1 as variance concentrates in a few directions. The *decision site* (or control-token position)

*Distiller Labs. Correspondence to Distiller Labs <orion@orionr.com>.

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is the assistant-header or end-of-prompt control token where the refusal gate and the judgment direction are defined; *content positions* are the token positions carrying the request text. The *moral subspace* is the rank-3 span of the moral mean-difference directions extracted from the moral-content datasets. The *refusal-decision* and *judgment-decision* directions are the mean-difference directions for refuse-versus-comply and for the moral judgment, read at the decision site. *Engage* and *disengage* name the two directions of a content intervention: engage adds harmful content, disengage removes it. Terms are defined again at first use below.

The six failure modes, with the section that treats each:

1. A projection-fraction instrument reads absence at a position where its own positive control has no discriminating power (the band-below-null decision site; Section 3.1).
2. A massive-activation outlier is a content-position statistic, so the decision-token bottleneck it seemed to contaminate is in fact clean (Section 3.2).
3. A covariance-matched null saturates in massive-activation families until every direction projects like a typical one (Section 3.3).
4. A per-head OV attribution overshoots the true residual write about threefold on reordered-norm architectures (Section 3.4).
5. A deliberation/prefill asymmetry statistic is operating-point-confounded when one arm sits at the ceiling (Section 5.1).
6. A cross-model asymmetry measured at the read layer is an artifact of measuring past the layer where one model already committed (Section 6).

Coverage is uneven across the panel: each mode is established on one or two models, not on all four, and the per-mode breakdown is in the Limitations section. Table 1 collects the participation ratios by model, position, and normalization so that a reader can tell which PR belongs where.

Mechanistic claims about model internals are claims about measurements. "Refusal reads the harm percept," "this head writes the refusal direction," "moral judgment is orthogonal to the refusal decision": each is a statement about a projection, a cosine, an ablation delta, or an interchange patch. The measurement can fail in ways that produce a clean-looking number. A covariance-matched null can saturate so that every direction projects like a typical one. A per-head attribution can overshoot the true residual write threefold. An interchange patch can go sign-chaotic because the outcome it reads is pinned at its ceiling. A "reads-X" verdict can be an artifact of measuring past the layer where the model already decided. None of these announce themselves; each returns a plausible scalar.

The discipline in this note was forced by one discovery. Across four architectures, the decision site (the assistant-header or end-of-prompt control token where the refusal gate and the judgment direction are defined) is a low-dimensional bottleneck. The participation ratio there is 14.7 on OLMo-3-7B-Instruct, 8.6 on Qwen2.5-7B, 10.2 on Llama-3.1-8B, and 12.8 on GPT-OSS-20B (a 20B reasoning MoE at its harmony decision token). A 9-to-15 effective-dimensional channel, on every model tested, while content positions at the same layers are full-rank-healthy (PR 40+/33+/35+). This is a substantive finding about where the refusal decision lives, and it belongs in the flagship. But it is also the reason the program's projection-fraction instruments failed: a positive control measured in a 15-slot channel projects onto its own span *less* than a random direction does, so the instrument had no discriminating power exactly where the interesting directions live. The finding and the failure are the same fact seen twice. This note carries the validity protocol the finding motivated; the flagship carries the finding.

(The bottleneck PR bar across the four architectures is **Figure 1**, which uses the in-format-ladder value 10.2 for Llama, comparable to OLMo 14.7 and Qwen 8.6; the decision-token measurement for Llama is 13.5, a second position and harness. Both are below 30.)

2 Related work

Each failure mode in this note has roots in an established line of work. We state the connection and what this note adds, so the reader can judge which cautions are new instruments and which are new framings of known ones.

Outlier dimensions and massive activations. That a few residual-stream dimensions carry a disproportionate share of variance, and that they distort geometric measurements, is well documented:

The decision site is a 9-to-15-dimensional control-token bottleneck (four open-weight models)

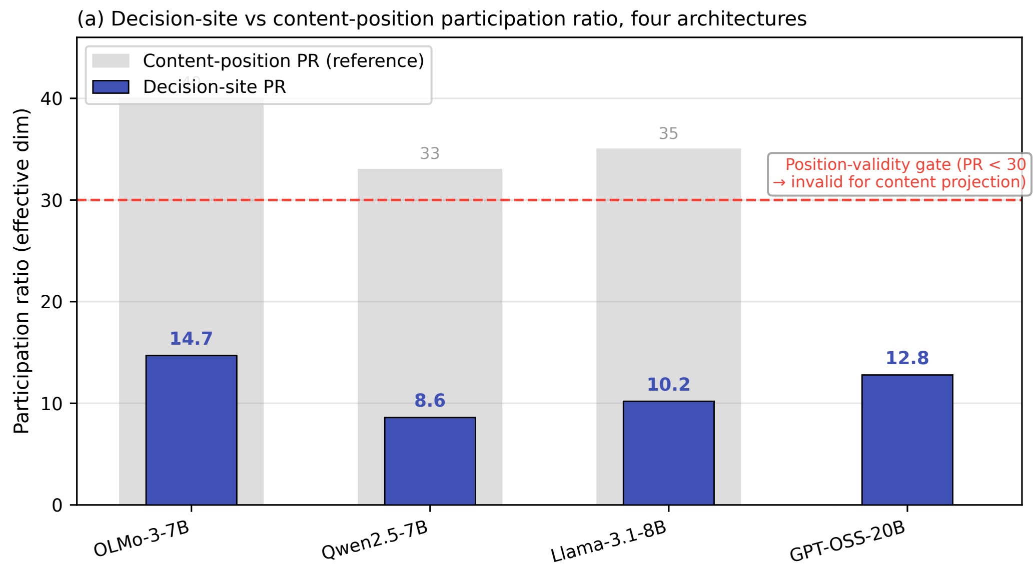

Figure 1: The decision-site participation ratio across four architectures: OLMo-3-7B-Instruct 14.7, Qwen2.5-7B 8.6, Llama-3.1-8B 10.2, and GPT-OSS-20B 12.8 (a 20B reasoning MoE at its harmony decision token). All four fall below the PR < 30 position-validity gate, while content positions at the same layers stay full-rank-healthy (PR 40+/33+/35+). The refusal decision lives in a 9-to-15 effective-dimensional control-token channel, on every model tested.

rogue dimensions obscure representational quality under cosine similarity (Timkey and van Schijndel, 2021), a small set of outlier dimensions disrupts transformers when removed (Kovaleva et al., 2021), and emergent outlier features appear at scale (Dettmers et al., 2022, Sun et al., 2024), related to the attention-sink phenomenon (Xiao et al., 2023). The standard response, per-dimension standardization, is not new here. What we add is that the *covariance-matched null* built for projection-fraction tests silently degenerates in these families: because the null draws random directions from the outlier-dominated covariance, every direction projects like a typical one, so the test loses discriminating power rather than returning an obvious artifact. Naming this instrument-level failure, and the band-below-null tell that catches it, is the contribution.

Norm handling in attribution. Reading a direction's per-head or per-layer contribution off the residual stream requires accounting for the block's normalization; the logit-lens and tuned-lens line makes the sensitivity to that choice explicit (Belrose et al., 2023). We add the specific, quantified failure for reordered (post-block) normalization as used by the OLMo-2/3 family: a naive per-head decomposition that skips the block norm overshoots the true residual write about threefold, and we give the exact RMSNorm-gain fold that recovers it, with a two-sided reconstruction gate (a one-sided floor misses overshoot).

Activation-patching methodology. That patching verdicts depend on metric, corruption, and layer choice is the subject of best-practice work (Zhang and Nanda, 2024). Our stimulus and depth sections are instances. An interchange readout run at a saturated outcome yields sign-chaotic deltas that mimic instrument failure, and a read-from verdict measured past the layer where the model has already committed can be a read-layer artifact. We frame both for reasoning models and supply the orthogonal-cell certificate and the commitment-relative depth as the checks.

Interpretability illusions. The general hazard, that a measurement can behave as if it found a mechanism when it has not, is the illusion literature: subspace activation patching can route through unintended pathways (Makelov et al., 2023), and individual-unit interpretations can be spurious (Bolukbasi et al., 2021). This note's thesis, that a broken instrument reads as a finding, is in that spirit. Our addition is the position-validity check, where a positive-control band that falls below the covariance null marks the measurement position itself as uninformative, and the integration of these checks into a pre-registration and verification protocol.

The scientific results that exercise these instruments are reported in a companion flagship study (in preparation); this note is the portable methodology, and its evidence is the model panel and single program it was derived on (see the limitations).

3 The decision-site instrument and its calibration ladder

Four failures converge on one object: the projection-fraction / cosine instrument used to ask whether a direction of interest lives inside a subspace. One is the position where it fails, another shows that position is architecture-general, a third is the null that degenerates underneath it, and a fourth is the attribution decomposition that overshoots when the same channel is read per-head. Each is stated as: failure $\rightarrow$ tell $\rightarrow$ protocol $\rightarrow$ certifying check.

Several participation ratios recur below, at different positions and under different normalizations. Table 1 lists them together so that each PR can be traced to its model, position, and normalization.

Table 1: Participation ratio (PR = $(\Sigma\lambda)^2/\Sigma\lambda^2$) by model, position, and normalization. The decision-site column is the in-format-ladder value plotted in Figure 1; the decision-token column is a second position and harness, measured for Llama only (13.5). Content-position PRs are full-rank-healthy. The geometric-cell column is the raw $\rightarrow$ standardized pair of Section 3.3, where per-dimension standardization lifts Qwen and Llama out of near-rank-1 collapse. GPT-OSS 12.8 is its harmony decision-token PR, treated as position-valid for the refusal decision-direction read against a separate MoE PR ceiling of 25 (Section 3.1). n/a marks a quantity not measured for that model.

Model	Decision-site PR	Decision-token PR	Content-position PR	Geometric-cell PR (raw $\rightarrow$ std)
OLMo-3-7B-Instruct	14.7	n/a	40+	43 $\rightarrow$ 94
Qwen2.5-7B	8.6	n/a	33+	1.0 $\rightarrow$ 39
Llama-3.1-8B	10.2	13.5	35+	1.5 $\rightarrow$ 89
GPT-OSS-20B	12.8	n/a	n/a	n/a

3.1 Band-below-null means the position is invalid, not that the direction is absent

Failure as it appeared. At the chat `final_pre_assistant` decision token on OLMo-3-Instruct, the positive-control moral band came out at [0.40, 0.47], and the honest covariance-matched null came out at 0.557. Held-one-out moral directions projected onto their own span *below* where random directions projected. Read naively, any direction of interest (refusal, judgment) that projected low there would read as "not in the moral subspace." That reading is unsupported: the instrument had no discriminating power at that position, so it cannot certify absence of anything.

The tell. The positive control sits below the null. Band-below-null $\Rightarrow$ position-invalid instrument. The moral band is not only a yardstick for "moral-adjacent"; it is a validity check on the measurement position. The cause here is dimensionality, not an outlier dimension (the top dim carries 0.2% of variance) and not a null that standardization can rescue (the null stays 0.52 after z-scoring). The channel is simply narrow: participation ratio 14.7. The $\sqrt{\texttt{(3/14.7)}}$ `= 0.45` heuristic (a rank-3 subspace at PR 14.7) predicts a median-scale projection; comparing that 0.45 against a null q95 of 0.557 and against the rank-3 pairwise-|cos| null of 0.41–0.51 is a consistency check that the numbers are the right size, not a convergence of three independent estimates on one value (0.45 is a median-scale prediction, 0.557 is a q95).

The protocol. `participation_ratio` is a required type-block field on every extracted direction, and any position with PR < 30 is flagged position-invalid for content projection-fraction tests at extraction time. All three chat decision sites (14.7 / 8.6 / 10.2) fall below the gate.

The certifying check and the reframe. Position-invalid does not mean uninterpretable model. A projection-fraction test fails there, but a decision-*direction* cosine does not: it is immune to the projection null. In a ~15-slot channel, refusal and judgment directions occupy different slots at

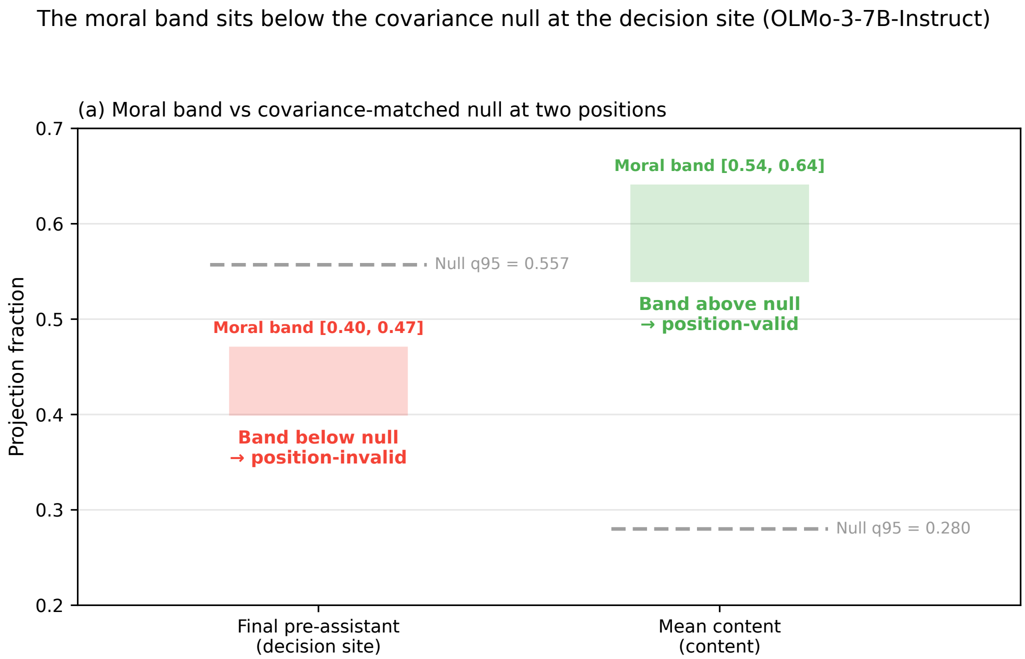

Figure 2: The calibrated ladder at the OLMo-3-Instruct chat decision token. The positive-control moral band [0.40, 0.47] sits *below* the covariance-matched null q95 of 0.557. A positive control below the null means the instrument has no discriminating power at that position for content projection-fraction tests: band-below-null implies the position is invalid, so a low projection there cannot certify absence. This is the visual form of the tell.

|cos| below even the low-dim random level, which reads as active separation, not a weak-instrument artifact. Concretely, refusal-decision is orthogonal to judgment-decision with no coupling detectable above |cos| 0.10 against a null q95 of 0.41 on OLMo (0.32 vs 0.42 on Qwen, 0.08 vs 0.51 on Llama). Geometrically, the moral-content band sits below the null at this bottleneck (band-below-null there, healthy at content positions), so content-versus-decision orthogonality is structurally favored here. This is a geometric observation, not a functional one: that moral content projects weakly onto the decision channel does not by itself establish that it fails to reach the decision, which is a causal claim that the note's own standard resolves only with an intervention cell. Read as geometry, any comprehension-to-decision coupling would have to ride the attention heads writing into the bottleneck, a concrete anatomical target.

One reconciling sentence is required for prose. The bottleneck is position-invalid for content projection-fraction tests (band-below-null) and position-valid for decision-direction reads (decision-direction cosine, and the GPT-OSS refusal projection). GPT-OSS's decision channel is called "position-valid (PR 12.8)" against a separate MoE PR sanity ceiling of 25; that ceiling is not the content rule.

Figure 2 is the calibrated ladder at this position: the moral band [0.40, 0.47] plotted below the covariance null 0.557, the visual form of the tell.

3.2 The massive-activation outlier is position-dependent, so the bottleneck is clean

Failure as it appeared. Llama-3.1 carries a massive-activation outlier: dim 788 holds 32% of residual variance. The worry was that this outlier contaminated every geometric read on Llama, including the decision-token cells.

The tell. The outlier's variance share is a *content-position* statistic. The decision token is a different position and had to be checked there, not assumed from the global number.

The protocol and check. At the decision-token channel where the refusal and judgment cells actually read, Llama is clean: participation ratio 13.5, covariance null 0.148, which barely moves to 0.114

under per-dimension standardization. The outlier lives at content positions, not at the ~13-dim control-token decision bottleneck, which is clean and low-rank across OLMo and Llama alike. So the "decision site is a narrow control-token channel" finding is cross-model, and the standardization fix matters more at content positions than at the decision token. This is why the null degeneracy (next) and the bottleneck are two different failures at two different positions, not one confound.

3.3 Covariance-matched nulls degenerate in massive-activation families

Failure as it appeared. The covariance-matched, rank-matched null (draw random directions from `N(0,` $\hat{\Sigma}$`)` of residual activations, project onto the rank-r subspace) is the honest null used throughout the program's earlier representational studies. On the instruct-model geometry it saturates: the moral-subspace projection null q95 = 0.92 on Qwen and 0.36 on Llama, the pairwise-cosine null q95 = 0.995 on Qwen and 0.90 on Llama, versus 0.26 on OLMo-3. At a saturated null every direction projects like a typical direction, so the test has no discriminating power.

The tell. The null value itself is near its ceiling. The mechanism is the same massive activations as the outlier finding above: Qwen dim 458 = 59% of residual variance, Llama dim 788 = 32%, OLMo-3's top dim = 1.4%. $\hat{\Sigma}$ is dominated by these dims, covariance-matched random directions nearly all align with them, and they project ~1 onto any subspace with a component there. The same dims collapse distinct raw mean-diff directions (Qwen ethics $\approx$ moral mean-diff $|\cos| = 0.90$). This is the known massive-activations / attention-sink phenomenon (Sun et al., 2024, Xiao et al., 2023).

The protocol. Recompute directions and the null in a per-dimension-standardized space (z-score by σ from a format/position-matched activation sample, sink tokens excluded), the primary fix. The criterion-based robustness variant projects out each dimension individually above 5% of variance. Behavioral results (ablation, judgment accuracy) never use this null and are untouched; only geometric cells need the re-audit.

The certifying check. The clean instrument must give the same verdict raw and standardized: OLMo, whose activations are well-conditioned, does. The quantitative before/after is the participation ratio (Table 1, geometric-cell column): raw PR = OLMo 43, Qwen 1.0, Llama 1.5 (one dim carries essentially all variance for Qwen and Llama); after z-scoring, PR = OLMo 94, Qwen 39, Llama 89. The raw PR $\approx$ 1 shows the collapse was near-total; standardization lifts Qwen and Llama into a genuinely multi-dimensional space.

A boundary case that names the residual limit. On the refusal-projection cell the two robustifications *disagree*: standardization gives refusal 0.20 above controls 0.10 (strong-form false), while top-k projection-out gives refusal 0.21 below controls 0.45–0.55 (strong-form true), and the same split appears on Llama. When standardization and projection-out disagree, the subspace is genuinely degenerate and needs a format or position change, not a null patch. The in-format chat ladder (whose decision-site space carries no >5%-variance dim, so it is outlier-free by construction) is the discriminator. This is the entry's own thesis applied to itself: no single null repair resolves a genuinely rank-1 space.

Scope of the fix. Which null a cell uses decides whether the degeneracy touches it. The instruct-model moral-subspace projection cells use the covariance-matched null (the one that degenerates); the program's Qwen/Llama geometric cells use a permutation test and raw (unnormalized) projection fractions; the behavioral cells use no geometric null at all. A companion audit found the permutation-and-raw-projection cells were never at risk: the permutation test's observed statistics are ~0.01 (unsaturated), the raw projection fractions are low and un-inflated (moral-subspace projection fraction 0.104 OLMo / 0.127 Qwen / 0.071 Llama, mean$|\cos|$ 0.04–0.07), and the moral-foundations subspace was built on the base model whose foundation directions did not collapse onto the outlier dim. The degeneracy is confined to the covariance-matched projection null applied to the instruct-model moral subspace. The general caution stands: covariance-matched nulls silently degenerate in massive-activation families, the field's default Llama/Qwen panel.

3.4 Reordered-norm architectures overshoot naive per-head OV attribution ~3$\times$

Failure as it appeared. The Stage-1 write attribution on OLMo-3-7B-Instruct (sum of per-head OV writes + per-layer MLP writes + embed onto the refusal direction, divided by the true residual write at the read layer) came back at 3.05. The linear decomposition overshot the actual residual write by 3×. The original gate (`recon ≥ 0.90`, one-sided) passed it, because a floor only catches undershoot.

The tell. A reconstruction well above 1.0 on a decomposition that should sum to 1.0. The mechanism is architectural: OLMo-2/3 use reordered (post-block) norm, applying `post_attention_layernorm` to the attention output and `post_feedforward_layernorm` to the MLP output *before* the residual add, with no input norm. The true residual write of the attention block is `RMSNorm(Σ_h W_O^h z_h)`, not the raw sum; the naive OV decomposition skips the norm, and since the raw block output has RMS above the norm's target it inflates ~3×. Pre-norm families (Llama, Qwen) write the raw block output to the residual and reconstruct ~1.0 natively, which is why the overshoot never appeared in Papers 1–7 (they used activations and directions, never OV decomposition).

The protocol. A two-sided gate `0.90 ≤ recon ≤ 1.10` (overshoot now fails), plus an exact RMSNorm fold. RMSNorm is diagonal at a fixed token, `norm(x) = (γ / rms(x)) ⊙ x`, so multiplying each pre-norm per-component write by the per-layer gain $g = \gamma \,/\, \mathrm{sqrt}(\mathrm{mean}(x^2) + \varepsilon)$ recovers the exact residual contribution. The fold fires automatically for reordered-norm models (detected via `post_feedforward_layernorm`) and is a no-op for pre-norm models.

The certifying check. The fold is exact: unit-tested to 1e-9, and it brings the Stage-1 reconstruction from 3.05 to 0.9999, inside the two-sided band. It affects only the head anatomy on OLMo-2/3 and other reordered-norm families (the un-folded numbers are inflated, for example the MLP write fraction was 0.23 un-folded and 0.384 folded). It does not touch the decisive causal cell, which reads the model's real forward pass with no decomposition. Per-head OV / logit-lens attribution silently overshoots ~3× on reordered-norm models unless the block norm is folded, a portable caution for a growing family (OLMo-2, OLMo-3, other post-norm designs).

4 Verdict discipline

Three estimator and intervention patterns gate how a number becomes a verdict.

4.1 Ratio-of-ratios over MDE-crossing

Whether an effect clears its minimum detectable effect is power-dependent. Comparing two effects by which side of the MDE each lands on is the overlap fallacy: it reads a difference in power as a difference in kind. Compare two effects instead by a within-outcome ratio and a bootstrap CI on the ratio difference (the estimator-traps trap-12 pattern).

Worked case: the `under_transfer` reclassification. The first headline was `reads_non_vmoral_features` at n=11, resting on an absolute transport comparison (a V_moral-restricted patch clears its MDE, a comparison patch does not). That absolute comparison was necessary but not sufficient. Re-run at n=23 with a within-outcome normalization, the honest verdict was `under_transfer`: the restricted patch moves refusal less than the full patch, but the two do not sit on opposite sides of a categorical line. The powered decisive cells (n=23 request-twins) are full→refusal −0.0833, V_moral-restricted→refusal −0.0282, complement→refusal −0.0636, harm-rank-1→refusal −0.0261, random-rank-3→refusal −0.0005, full→judgment +0.0459, restricted→judgment +0.0237, against a refusal MDE of 0.0238 and a judgment MDE of 0.0086. `under_transfer` was then itself superseded by the rank sweep (`harm_saturating`, Section 7), but the estimator lesson is the one that recurs: the reclassification from `reads_non_vmoral_features` to `under_transfer` happened because the ratio, not the MDE-crossing, is the comparison of record. (The specificity claim that survives is stated as a difference CI, not an overlap check: V_moral-restricted moves refusal more than a random rank-3, Δ = 0.031, paired 95% CI [0.020, 0.043], excludes 0.)

A second instance, from reasoning models: an early raw diff-of-means null (harm direction 0.44–0.49 of residual norm) read as "harmfulness is not causally encoded," but that was a magnitude artifact. Reply-inversion recovered the causal signal (Qwen2.5-14B-Instruct shift +17.4 flips 33%, Llama-3.1-8B-Instruct +3.0 flips 23%). Magnitude and residual-norm share are not causal relevance; a causal readout is.

4.2 Power tables before compute

Compute MDE(n) from measured within-condition variance before spending compute. If no feasible n resolves the effect, the block is the instrument, not the sample, and the compute run is futile.

Worked case: the Llama bounded-unresolved table. The Llama refusal cells came back chaotic (Section 4.3, Section 5). The temptation was a larger same-design re-run. The power table, built from saved within-level arrays, said the re-run was futile: the ratio-of-ratios CI on the latched denominator was $[-2.3, 4.9]$, and no feasible n at that variance closes it, because the denominator is saturated rather than noisy. The underpowered Llama `R_refusal_k` and `R_judgment_{k>1}` cells were voided as denominator-latched, and the clean channel was identified as the *reverse* (disengage) direction, not more samples in the forward one. An afternoon of saved-array work prevented a compute run. This is the general rule: saving per-pair / per-rollout / per-head arrays by default keeps the power computation zero-GPU, so futility is caught before the session, not after.

4.3 The orthogonal-cell certificate

When a causal readout comes back chaotic, root-split against an orthogonal outcome the same intervention should move. If the orthogonal cell is coherent, the instrument is certified and the chaos is a property of the read-out outcome, not a broken patch.

Worked case: Llama content-swap. On Llama the content-swap interchange patch produced sign-chaotic refusal deltas (SD 0.31, median +0.029) against OLMo's clean −0.083, which read as a broken instrument. The judgment cell is the positive control: the *same* patch moved judgment coherently (CI excludes 0). So the patch works. The refusal chaos is saturation: the boundary-violating twins sit at the refusal ceiling (baseline refuse 0.83–1.0), so the decision-token refusal projection is latched and has no room to move, and the refusal-delta SD grows with severity (0.296 → 0.352) as saturation deepens. OLMo's refusal moved because it was weak (unsaturated). A causal readout run at a saturated outcome yields chaotic, sign-unstable deltas that mimic instrument failure; the orthogonal cell tells the two apart.

5 Stimulus discipline

The operating-point rule: discrimination screens must bracket the point where the outcome actually moves. A saturated outcome latches the readout. Use a severity ladder and a boundary band (outcome ~0.5), and report the psychometric curve, not a single point.

This is where the readout itself can lie. Reasoning models defeat clean judgment readouts (regex, final-answer, forced-logit), while a plain instruct model is clean on the same battery (Qwen2.5-14B-Instruct: 24/24 harmless-safe, 24/24 harmful-harmful). The readout has to be validated on the model class it is run on, at an operating point where the outcome is not pinned.

5.1 A deliberation/prefill asymmetry is operating-point-confounded when the gate is a step

Failure as it appeared. On GPT-OSS-20B the reasoning-prefill deliberation cell (engage = inculpating prefill, disengage = exculpating prefill) emitted a clean-looking asymmetry `A = 1.0`: engage flips benign→refuse, disengage flips violating→comply 0/7. It read as one-way early commitment, as if deliberation only ever pushes toward refusal.

The tell. `A = 1.0` with a bootstrap CI of width 0. The disengage arm is uniformly 0, so every resample returns 1: a degenerate CI, not a precise estimate. The rule-of-three exposes it directly, disengage 0/7 gives a 95% upper bound of ~0.43, not 0. The asymmetry is the same dynamic-range confound as Section 4.3: the disengage arm was tested on violating items that already refuse at baseline (at the ceiling), while the engage arm was tested on unsaturated benign items (with room to move up). An asymmetry statistic that compares an arm-with-headroom against an arm-at-the-ceiling measures the operating points, not the model. A companion trap in the same run: the harm-separability commitment curve reads ~1.0 from the first trace bin, but that measures when *harm is represented* (harmful and harmless traces differ from the start), not when the *decision* is fixed.

Why the usual fix is not enough here. That fix was boundary-band twins (outcome ~0.5). GPT-OSS's gate is a step: the severity ladder finds no unsaturated violating level (empty boundary band, 5.6% of items in the mid-band). The operating point cannot be bracketed behaviorally at the existing resolution, so more boundary-band stimuli do not exist to collect.

The protocol (the de-confounder). Replace the binary disengage flip with a graded exculpatory prefill series (weak→strong) and read a continuous projection (the decision-channel residual under each prefill onto the refusal direction) alongside the behavioral flip. The graded projection registers sub-flip movement, so "no flip at maximum prefill" splits cleanly into *reversible* (projection moves toward comply) versus *genuine downward-robustness* (projection flat). Saturation can no longer masquerade as commitment. A pre-registered band-existence check decides whether a finer ladder is even buildable: the per-item base-refuse histogram is read as smooth (resolution-limited, build a finer ladder) versus bimodal (a step, switch to the graded readout).

The certifying check. These GPT-OSS behavioral cells are small, n = 7 to 10 items per arm, and the flip fractions below should be read at that sample size. Under the graded series, GPT-OSS is a reversible reader: strong exculpatory prefill flips ceiling-refusing violating items to comply 6/10 (reported without a CI at n=10), and the decision-channel refusal projection moves monotonically toward comply in all 10 items (frac_projection_moved 1.0, frac_monotone 1.0). The behavioral flip is the primary readout; the monotone projection corroborates it (with a last-token caveat on the projection). The engage direction is separately consequential: an inculpating-analysis prefill flips unsaturated benign requests to refuse 7/7 (Wilson 95% [0.65, 1.0]), so the decision is not fixed before the trace. The first-run disengage 0/7 was the saturation trap, now resolved. Report the behavioral flip and the graded readout separately; the asymmetry statistic itself is not reported, because it is uninterpretable when one arm sits at the ceiling.

This is the operating-point rule's reasoning-model instance: when the gate is a step and the boundary band is empty, do not report the asymmetry; switch to a graded intervention with a continuous readout that registers sub-threshold movement.

6 Depth discipline

A verdict about what a circuit *reads* must state the intervention depth relative to the model's commitment. A patch at the read layer is post-commitment for an early-committing model, so a "reads-X" or asymmetry claim measured there can be a read-layer artifact. Measure at (and report) the pre-commitment coherent depth, and depth-match cross-model comparisons.

Failure as it appeared. The naive cross-model asymmetry at the read layer (layer 16) was `A_Llama = +0.82`, engage-dominant and latch-like, against `A_OLMo = −0.20`, for a difference of +1.03 (95% CI [0.16, 1.61], excludes 0). Read at face value, this said Llama's refusal has a third distinct property, a hard directional latch that OLMo lacks.

The tell. Llama's disengage is coherent at earlier layers but not at the read layer. Patch-layer sweep: Llama's disengage is coherent at layers 8/12/14 (−0.12 / −0.11 / −0.20, CIs exclude 0) and incoherent at layer 16 (−0.014). Llama commits *before* the read layer; OLMo's disengage is coherent at layer 16 (−0.62), so OLMo commits at or after it. Measuring Llama's asymmetry at layer 16 measures it after Llama has already decided, so the +0.82 is a post-commitment artifact. (The two layer-12 disengage numbers are two cells: the patch-layer sweep above reads −0.11, the depth-matched full re-run reads −0.57, both coherent; the verdict is the same.)

The protocol. Depth-match the comparison to the pre-commitment coherent layer, layer 12, and recompute both models there.

The certifying check. At matched layer 12, `A_Llama = −0.28` (CI [−0.47, +0.03]) and `A_OLMo@12 = −0.54` (CI [−0.81, −0.32]), a difference of +0.26, down from +1.03 at the read layer. The Llama asymmetry CI [−0.47, +0.03] includes 0, and no difference-CI is reported on the residual +0.26, so the surviving cross-model claim rests on the reads-axis difference, not on a residual asymmetry. The apparent third property collapses: the asymmetry is a *consequence* of early-commitment, not an independent latch. What survives depth-matching is the reads-axis difference: at layer 12 Llama reads broad (`broad_moral`: R_refusal 0.85 ≈ R_judgment 0.79, gap closes, harm-rank-1 only 0.59) while OLMo stays harm-keyed (R_refusal 0.43 < R_judgment 0.53, gap open). The read-layer +0.82 is retained only as a voided number with its replacement, never as a finding.

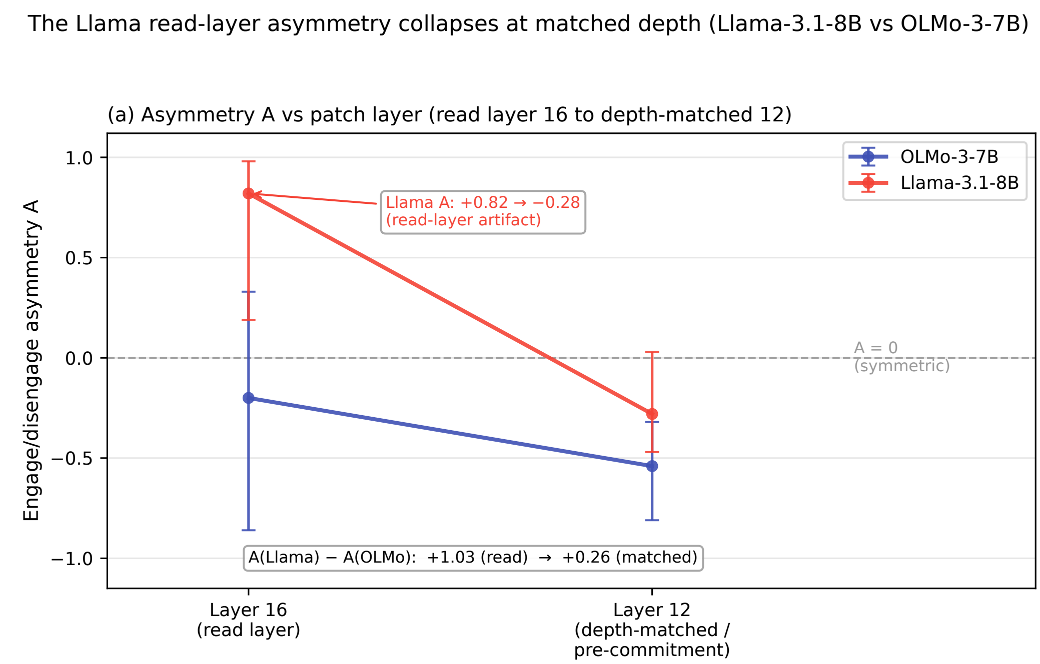

Figure 3: Depth collapse of the cross-model asymmetry A. Measured at the read layer (16), A_{Llama} is $+0.82$ and reads as a Llama-specific directional latch. Depth-matched to the pre-commitment coherent layer (12), it falls to -0.28, while A_{OLMo} moves from -0.20 to -0.54; the cross-model difference collapses from $+1.03$ to $+0.26$. The apparent third property is a read-layer artifact of measuring after Llama has already committed.

Figure 3 is this depth collapse: `A_Llama` from +0.82 at the read layer to -0.28 at matched layer 12, with `A_OLMo` $-0.20 \to -0.54$, the depth-indexed exemplar.

7 Case study: one program's amendments as caught failures

The program asked two questions of the refusal decision: *what* moral content it reads, and *how* it commits. Its pre-registration trail is a sequence of caught failures, each a dated amendment. Read in order, it is the methods note in miniature. The public amendment trail is a credibility asset; it is cited from the flagship, not hidden.

- **Null degeneracy.** The instruct-model covariance null saturated on Qwen/Llama. Fix: standardized recompute; OLMo unchanged raw→standardized certified it.
- **Position gate.** The decision site is a PR-14.7 bottleneck with the positive control below the null. Fix: PR<30 position-invalid flag; V_moral re-typed as format-robust (invalid-position artifact at `final_pre_assistant`, band matches at the valid `mean_content` position), and the content-projection numbers re-typed as non-verdict.
- **Referee-pass hardening.** Before any asset was built, a referee pass re-typed the twin stimulus (request-twins carrying the judgment outcome, Δrefusal expected-flat), added a transport positive control to the decisive cell, made the head-score null channel-matched (mean/resample ablation, not zeroing), and added a behavioral-discrimination pilot screen.
- **OV overshoot.** Per-head attribution reconstructed at 3.05 on OLMo-3's reordered norm. Fix: two-sided gate + exact RMSNorm fold, reconstruction 3.05 → 0.9999.
- **MDE-crossing headline.** The `reads_non_vmoral_features` verdict (n=11) rested on an absolute transport comparison. Fix: within-outcome ratio at n=23 → the honest `under_transfer`.
- **Under-transfer superseded.** A rank sweep replaced the point comparison: as k ∈ {1, 3, 8, 16}, R_judgment climbs 0.05 → 0.46 → 0.59 → 0.66 while R_refusal saturates 0.01 → 0.31 → 0.26 → 0.27 at the harm-rank-1 level (harm_rank1_R 0.31), random-null ~0 at every rank. The one-knob model `R_refusal(k)` ≈ `min(harm_ceiling, R_judgment(k))` fits the

plateau ($k \geq 3$) at RMSE 0.036, and PC1 (highest variance, purity 0.974, most harm-aligned at cos 0.35) is causally inert (rank-1 moves neither readout, 0.01 / 0.05), the lesson that variance is not causal relevance. Both the one-knob RMSE 0.036 and the PC1-inert reading are illustrative point estimates, reported without CIs. Verdict: `harm_saturating`.

- **GPT-OSS commit axis.** The first session banked the position gate (PR 12.8), consequential engage deliberation (benign→refuse 7/7), and the first-run disengage 0/7 that looked irreversible.
- **Power table.** The saved-array power computation ruled the Llama same-design re-run futile before the compute run (ratio-of-ratios CI [-2.3, 4.9] on a latched denominator).
- **One-root diagnosis.** The Llama chaos was diagnosed by a single root split, judgment-delta coherence, not a grab-bag of probes: the orthogonal judgment cell is coherent, so the refusal chaos is saturation.
- **Denominator-latched voids.** The Llama "reads beyond harm" hint (R_refusal 0.44 vs harm-rank-1 0.14 at rank 16) was voided, its denominator saturated; the three branches re-entered unweighted and were resolved by the depth-matched `broad_moral` read.
- **Nomenclature + early-commitment.** Fixed engage = harm-add / disengage = harm-remove; defined the asymmetry statistic A; the patch-layer sweep gave the EARLY-COMMITMENT verdict (Llama disengage coherent at 8/12/14, incoherent at 16) and the read-layer cross-model asymmetry A_Llama − A_OLMo = 1.03 (later depth-re-attributed, Section 6).
- **Depth-indexed verdict.** The +0.82 read-layer asymmetry collapsed to -0.28 at matched layer 12 (Section 6). This amendment started this note.
- **Harm-coextensive hardening.** The reads-broad verdict survived the rank-1 harm-coextensive alternative: a single harm cue spans only 3.6% of the engage-driving moral basis (the rank-2/4 severity-ladder version is a stated extraction rider on unsaved contrasts).
- **Graded disengage.** The step-gate saturation trap was de-confounded: GPT-OSS is a reversible reader, violating→comply 6/10 with monotone projection in all 10 items (Section 5.1).
- **Confound-named hypothesis.** The n=3 categorical co-occurrence ("harm-readers reversible, broad-reader early-commits") was replaced by a falsifiable dimensionality→reversibility hypothesis with an explicit architecture confound: the read↔commit pairing is confounded by lineage/scale/tokenizer/reasoning-vs-instruct at three points, and is deconfounded only by varying one axis at a time. The measured two-axis table stands; its interpretation is a follow-on hypothesis, not an n=3 claim.

7.1 Reflexive discipline: the program audits its own published paper

A cold-boot re-read turned the same scrutiny on the program's own published work. Paper 1 (Reblitz-Richardson, 2026, arXiv:2606.11375v1, 9 Jun 2026) stated a raw layer-depth fragility gradient as its abstract-level Finding 2: late layers were reported as far more fragile than early ones, with a raw late/early σ^* ratio up to ~14.7× (the claim ledger records the range as 7–15×; Table 2 late 10.0 / early 1.8), plus a raw post-saturation σ^* decline from 18.3 to 4.7. A post-submission control (§4.4, RMS normalization) shows the gradient is largely an activation-scale artifact: under RMS normalization the ratio collapses to ~1.8–2× (the residual ~2× is not claimed as a genuine gradient, since RMS controls scale not covariance shape), the cross-checkpoint ordering fails at 8/37 checkpoints, and the post-saturation decline is withdrawn (flat, ~13.8 → 15.0). The lesson is exact: raw σ^* is valid within-layer (same activation scale) but activation-scale-confounded cross-layer; RMS-normalize for any cross-layer claim.

Two things about *how* it was caught belong in this note, and they are not the same thing. First, what caught the error was a cold-boot (fresh-context) ledger re-read after publication, not the calibration protocol firing at authoring time: the confound surfaced when a fresh-context audit re-read a result the warm working sessions had produced and re-read many times without flagging. The fresh-context reviewer's advantage is real, and mechanically recreating it caught an abstract-level error. That the written calibration protocol would have flagged this prospectively is a separate and weaker claim the note does not establish; the honest record is that a published number stood until a fresh re-read caught it. Second, it triggers a v2 erratum on a published paper. A program that runs an instrument-calibration discipline on other people's panels has to run it on itself; the same scale confound named in the covariance null (magnitude is not the signal) is the one that inflated Finding 2. This is the reflexive instance, and it is the reason the note leads with "instruments before verdicts" rather than presenting the direction results as settled.

7.2 Claim hygiene: a traceable claim ledger

Every number in the program traces to an anchored-sentence row in a claim ledger: if a draft states a scalar that ledger does not carry, the draft is wrong until a row is added. The ledger also carries a register of superseded claims retained *with their replacements*, so they cannot re-enter prose as findings. The three the reader of a naive draft would most likely resurrect are all there: the un-folded 3.05 head anatomy (replaced by the folded 0.9999), the n=11 `reads_non_vmoral_features` headline (replaced by `under_transfer` at n=23), and the `A = +0.82` read-layer asymmetry (replaced by the depth-matched −0.28). A separate set of number-integrity flags blocks specific *scalars* whose value is still in dispute across documents, without blocking the verdicts (the shapes and signs are robust; only a printed number waits on the flag). Voided results may be discussed as methods lessons in this note; they are never findings in the flagship.

8 Limitations

The discipline in this note is drawn from one research program, and its scope should be read accordingly.

Per-mode model coverage. The summary "six failures across four architectures" is honest only about the panel as a whole; each individual mode is established on one or two models, not on all four. The band-below-null position-invalid instrument (Section 3.1) is shown on OLMo-3-Instruct at its decision token, with the $PR < 30$ gate applied on OLMo, Qwen, and Llama. The massive-activation outlier's position-dependence (Section 3.2) is a Llama-3.1 finding cross-checked against OLMo. The covariance-matched null degeneration (Section 3.3) is a Qwen-and-Llama result with OLMo as the clean control. The reordered-norm OV overshoot (Section 3.4) is OLMo-only, since pre-norm Llama and Qwen reconstruct near 1.0 natively. The deliberation/prefill asymmetry (Section 5.1) is GPT-OSS-only. The read-layer depth artifact (Section 6) is Llama-versus-OLMo. So the note is six protocols, each demonstrated on one or two members of a four-model panel, not six effects each seen on four models.

The position-validity gate can flag a real direction. The $PR < 30$ gate declares a position invalid for content projection-fraction tests, but its false-invalid rate is unquantified. A genuine content direction present at a narrow position would be flagged the same way as a weak-instrument artifact. The gate is calibrated to catch band-below-null cases; it is not calibrated against a bank of known-present directions at narrow positions, so it can suppress a real read.

Standardization can destroy anisotropic signal. Per-dimension standardization rescues the covariance-matched null in massive-activation families (Section 3.3), but z-scoring flattens genuine anisotropy along with the artifactual kind. The note's own boundary case, where standardization and top-k projection-out disagree on the same cell (Section 3.3), is the symptom: when a subspace is genuinely low-rank, standardization can report structure that projection-out does not, with no guarantee the standardized space preserves a real anisotropic direction.

Survivorship bias. Only caught failures appear here. The note has no estimate of how many measurement failures the discipline missed, and a protocol that reports only its catches cannot state its own false-negative rate. The six are the ones a fresh-context re-read or a positive control happened to surface; failures that produced a plausible number and were never re-examined would not be in this note by construction.

Single program, single panel. All of the evidence is from one causal interpretability program on one model panel (OLMo-3, Qwen2.5, Llama-3.1, GPT-OSS). The protocols are offered as portable within that program and worth testing elsewhere, not as externally validated across programs, tasks, or architectures beyond this panel.

9 Checklist: the reusable protocol

The ship-blocker gates below are the portable form of the program's discipline. They are appendix-form restatements of the companion skills.

Before any projection-fraction / cosine geometric cell:

- Record `participation_ratio` at the measurement position. If PR < 30, the position is invalid for content projection-fraction tests; report a decision-direction cosine instead, or move to a valid position.
- Check the positive-control band against the covariance null at that position. Band-below-null means the instrument has no discriminating power there; do not report absence.
- In a massive-activation family (any Llama/Qwen-class panel), inspect the null value for saturation and the top-dim variance share. Standardize (z-score, sinks excluded) and, as a robustness variant, project out each >5%-variance dim. Certify with a clean model (raw$\rightarrow$standardized same verdict). When standardization and projection-out disagree, the space is genuinely degenerate; change format or position, not the null.

Before any per-head OV / logit-lens attribution:

- Detect reordered norm (post-block `post_feedforward_layernorm`). If present, fold the per-layer RMSNorm gain (unit-test the fold to ~1e-9). Gate reconstruction two-sided ($0.90 \leq$ recon ≤ 1.10); a one-sided floor misses overshoot.

Before any patch / ablation / interchange / steering verdict:

- Pre-register the intervention spec block (stimulus–outcome baseline matching, transport positive control, channel-matched specificity null, token-alignment rule, harness parity for the outcome classifier) before the run.
- Certify a chaotic readout with an orthogonal cell the same intervention should move; if the orthogonal cell is coherent, the chaos is saturation, not a broken instrument.
- Compare two effects by a within-outcome ratio + bootstrap CI on the ratio difference, never by which side of the MDE each lands on.
- State the intervention depth relative to commitment; depth-match cross-model comparisons to the pre-commitment coherent layer.

Before any null / orthogonality / below-threshold verdict:

- Build the calibrated ladder (floor $\rightarrow$ matched null $\rightarrow$ measurement $\rightarrow$ positive band) and re-verify every control's defining property in the current context.
- State the minimum detectable effect. A null without an MDE has no teeth; write the detection bar into the sentence ("no coupling detectable above |cos| 0.10 against a null q95 of 0.41," not "dissociation").

Before any stimulus screen:

- Bracket the operating point with a severity ladder and a boundary band (~0.5); report the psychometric curve. If the gate is a step (empty boundary band), switch to a graded intervention with a continuous readout and report the behavioral flip and the graded readout separately. Validate the readout on the model class it is run on.

Before a compute run:

- Compute MDE(n) from measured within-condition variance. If no feasible n resolves the effect, the block is the instrument, not the sample; do not spend the compute. Save per-pair / per-rollout / per-head arrays by default so the power computation and later statistics stay zero-GPU.

Before any commit or draft:

- Every printed scalar traces to an anchored claim-ledger row; voided numbers stay in a register of superseded claims with their replacements so they cannot re-enter prose as findings.

Reproducibility

Each figure in this note ships with a regeneration script that reads a committed CSV under the convention `papers/figure_data/mn_*.csv` (`mn_bottleneck_pr.csv`, `mn_ladder.csv`, `mn_depth_collapse.csv`); the analysis outputs are gitignored, so the committed CSV plus its script is the reproducibility contract for every figure.

Every number in this note, the participation-ratio profiles, the calibrated covariance nulls, the positive-control ladders, the per-head write attribution, and the rank-sweep outcomes, is indexed in the shared supplement `deepsteer/supplement/MANIFEST.json`, each with a content hash, its provenance, and the figure or table it backs. The two instruments this note shares with the companion flagship (the decision-site participation-ratio profile and the depth-asymmetry panel) live in the supplement once and are cited by both papers; `deepsteer/supplement/scripts/verify.py` asserts the note's plotting copies match the canonical values, so a shared number can change in only one place. Model ids, decision layers, standardization settings, and seeds are pinned in `deepsteer/supplement/PROVENANCE.md`; raw activation caches are available on reviewer request.

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